

Data Science Unit1 Question Bank

1. Define Data Science. [2 Marks]

Data science is a deep study of the massive amount of data, which involves extracting meaningful insights from raw, structured, and unstructured data that is processed using the scientific method, different technologies, and algorithms.

2. Explain reasons to use data science technology. [5 Marks]

- Convert the massive amount of raw and unstructured data into meaningful insights.
- Many companies like Google, Amazon, Netflix, etc. opt data science for better customer experience.
- Data science can help in different predictions such as various survey, elections, flight ticket confirmation, etc.
- Industries can quickly examine their problems and successfully address them using data science.
- Products and businesses can better connect with their customers when they use this data to tell their stories to viewers.
- Social Media analytics can be done with help of data science.

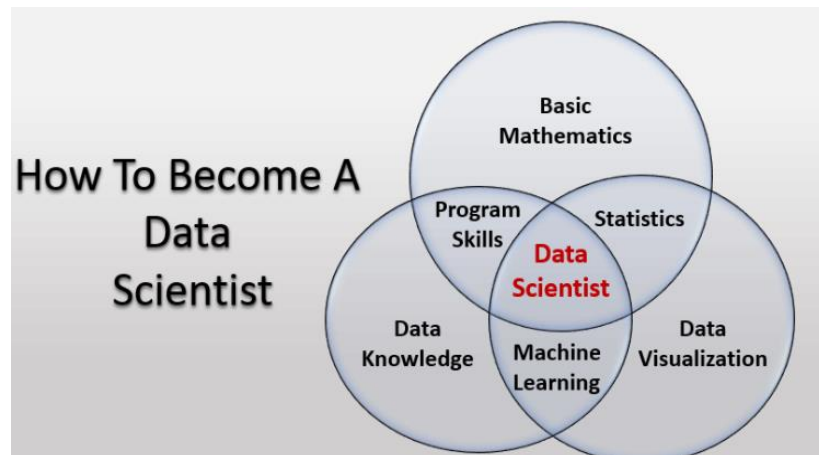
3. Define Data Scientist. [2 Marks]

- A data scientist is an analytics professional who is responsible for collecting, analysing and interpreting data to help drive decision-making in an organization.
- Professionals who can dig in and find valuable business insights

4. What is the job description of a data scientist? [5 Marks]

- Fetching data from valuable data sources.
- Processing, cleansing, and validating the integrity of data to be used for analysis
- Analysing large amounts of information to find patterns and solutions
- Developing prediction systems and machine learning algorithms
- Presenting results in a clear manner
- Propose solutions and strategies to tackle business challenges
- Collaborate with Business and IT teams.

5. What are the key skills required to become data scientist? [5 Marks]



- Programming Skills – knowledge of statistical programming languages like R, Python, and database query languages like SQL, Hive, Pig is desirable. Familiarity with Scala, Java, or C++ is an added advantage.
- Statistics – Knowledge of statistical tests, distributions, regression, maximum likelihood estimators, etc.
- Machine Learning – good knowledge of machine learning methods like k-Nearest Neighbours, Naive Bayes, SVM, Decision Forests.
- Strong Math Skills (Multivariable Calculus and Linear Algebra) - understanding the fundamentals of Multivariable Calculus and Linear Algebra is important as they form the basis of a lot of predictive performance or algorithm optimization techniques.
- Data Wrangling – proficiency in handling imperfections in data is an important aspect of a data scientist job description.
- Experience with Data Visualization Tools like matplotlib, ggplot, d3.js., Tableau that help to visually encode data
- Excellent Communication Skills – it is incredibly important to describe findings to a technical and non-technical audience.

6. List various applications of data science. [5/10 marks] any 5 for 5 marks

Healthcare

Healthcare companies are using data science to build sophisticated medical instruments to detect and cure diseases.

Gaming

Video and computer games are now being created with the help of data science and that has taken the gaming experience to the next level.

Image Recognition

Identifying patterns is one of the most commonly known applications of data science. In images and detecting objects in an image is one of the most popular data science applications.

Recommendation Systems

Next up in the data science and its applications list comes Recommendation Systems. Netflix and Amazon give movie and product recommendations based on what you like to watch, purchase, or browse on their platforms.

Logistics

Data Science is used by logistics companies to optimize routes to ensure faster delivery of products and increase operational efficiency.

Fraud Detection

Fraud detection comes the next in the list of applications of data science. Banking and financial institutions use data science and related algorithms to detect fraudulent transactions.

Internet Search

Search engines, such as Google, Yahoo, etc employ data science algorithms to offer the best results for our searched query in a matter of seconds. Given that Google handles more than 20 petabytes of data per day.

Speech recognition

Speech recognition is one of the most commonly known applications of data science. It is a technology that enables a computer to recognize and transcribe spoken language into text. It has a wide range of applications, from virtual assistants and voice-controlled devices to automated customer service systems and transcription services.

Targeted Advertising

If you thought Search was the most essential data science use, consider this: the whole digital marketing spectrum. From display banners on various websites to digital billboards at airports, data science algorithms are utilised to identify almost anything.

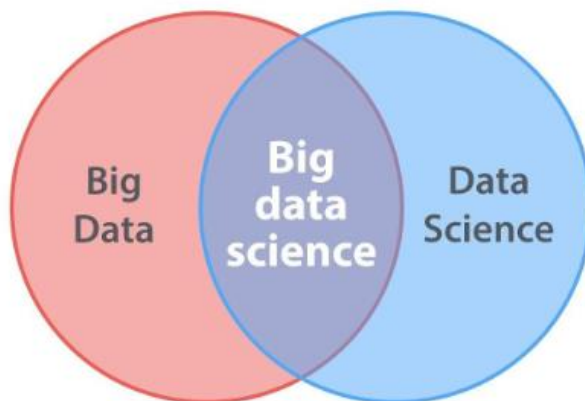
Airline Route Planning

Next up in the data science and its applications list comes route planning. As a result of data science, it is easier to predict flight delays for the airline industry, which is helping it grow.

7. Define Big data. Give example.

- The term “big data” refers to data that is so large, fast or complex that it’s difficult to process using traditional methods.
- Example: Amazon, Facebook, Google data

8. How Big data and data science work together.[5 Marks]



- Data Science uses scientific methods, processes, algorithms and systems to extract insights and knowledge from data to generate enterprise value.
- Big Data is about processing data in large volume, variety, velocity.
- Big Data is part of data Science .
- Big Data and Data Science are complementary to each other or even needed together to achieve better dynamics for the analytics in Big Data Analytics results, or to supply Data Science with a larger data amount.
- **Therefore, one can call the combination of Big Data and Data Science simply Big Data Science**

- Data Science enables companies to efficiently understand big data from multiple sources.
- Example: Face book analysis dashboard.

9. Why Python for data science? [5 Marks]

- **Easy to Learn:** Being an open-source platform, Python has a simple and intuitive syntax that is easy to learn and read. This makes it a great language for beginners to learn data science.
- **Cross-Platform:** Being a developer, you don't need to worry about the data types. The reason is, Python allows developers to run the code on Windows, Mac OS X, UNIX, and Linux.
- **Portable:** Being an easy & beginner's friendly programming language, Python is highly portable in nature which means that a developer can run their code on different machines without making any further changes.
- **Extensive Library:** Python has several powerful libraries that make data analysis and visualization easy. Pandas is a library for data manipulation and analysis, NumPy is a library for numerical computation, and Matplotlib is a library for data visualization.
- **Community Support:** Python has a large and active community that supports and contributes to the development of various libraries and tools for data science. This community has created many useful libraries, including Pandas, NumPy, matplotlib, and SciPy, which are widely used in data science.

10.What is Data Analytics? [2 Marks]

- **Data analytics** is the process of analysing raw data in order to draw out meaningful, actionable insights, which are then used to inform and drive smart business decisions.
- Data is extracted from various sources and is cleaned and categorized to analyse various behavioural patterns.

11.List types of data analytics [2 Marks]

- Descriptive analytics,
- Diagnostic analytics,
- Predictive analytics
- Prescriptive analytics.

12. Define Data Analytics. Explain types of data analytics. [5 Marks]

- **Data analytics** is the process of analysing raw data in order to draw out meaningful, actionable insights, which are then used to inform and drive smart business decisions.
- **Descriptive analytics**
 - helps answer questions about what happened.
 - These techniques summarize large datasets to describe outcomes to stakeholders.
- **Diagnostic analytics**
 - Helps answer questions about why things happened.
 - These techniques supplement more basic descriptive analytics.
- **Predictive analytics**
 - Helps answer questions about what will happen in the future.
 - These techniques use historical data to identify trends and determine if they are likely to recur.
- **Prescriptive analytics**
 - Helps answer questions about what should be done. By using insights from predictive analytics,
 - This allows businesses to make informed decisions in the face of uncertainty.

13. List and explain areas of data analytics. [5 Marks] any 5

- **Policing/Security**
 - Data analytics application will make us have safer cities without police putting their lives at risk.
- **Transportation**
 - The TFL and train operators made use of data analytics to ensure the large numbers of journeys went smoothly. They were able to input data from events that took place and forecasted a number of persons that were going to travel
- **Fraud and Risk Detection**
 - Banks learning to divide and conquer data from their customers' profiles, recent expenditure and other significant information that were made available to them. This made it

easy for them to analyse and infer if there was any probability of customers defaulting.

- **Manage Risk**
 - Analytical software is used for detecting the various forms of fraudulent claims. Risky claims are detected by red flag indicators which can be examined.
- **Customer Interaction**
 - To improve the overall experience of customers, it is best for insurance companies to provide a wide range of communication methods for their customers.
- **City Planning**
 - Data analytics applications, as well as modeling, would make it easy to mark the outcome of erecting a structure on any spot.
- **Healthcare**
 - The costs of medical treatments are extremely high. While the cost of prevention is vastly lower. We're talking about customers, health plans, hospitals, and clinics.
- **Travel**
 - Data analytics applications help in the optimization of traveler's buying experience via social media and mobile/weblog data analysis
- **Internet/Web Search**
 - search engines such as Google, Bing, Yahoo etc. is a result of data science applications because they use algorithms to deliver the best results for any search query directed at them in just a split second.
- **Digital Advertisement**
 - Apart from web search, there is another area where data analytics and data science serves a very important purpose – digital advertisements. From the banners displayed on several websites to the digital billboards seen in the big cities; all are controlled by data algorithms.

14. List any 4 tools used for Data Analytics. [2 Marks]

- R Programming
- Tableau

- Python
- SAS
- Apache Spark
- Excel

15.Explain data analytics tools. [5 Marks]

R Programming

- It can easily manipulate your data and present in different ways.
- R compiles and runs on a wide variety of platforms UNIX, Windows and MacOS.
- R also provides tools to automatically install all packages as per user requirement, which can also be well assembled with Big data.

Tableau

- Tableau Public is a free software that connects any data source be it corporate Data Warehouse, Microsoft Excel or web-based data.
- Creates data visualizations, maps, dashboards etc. with real-time updates presenting on web. They can also be shared through social media or with the client.

Python

- Python is an object-oriented scripting language which is easy to read, write, maintain and is a free open source tool.
- Wide range of packages for analysis and visualization.

SAS (Statistical Analysis Software)

- SAS is easily accessible, manageable and can analyse data from any sources.
- SAS introduced a large set of products in 2011 for customer intelligence and numerous SAS modules for web, social media and marketing analytics.

Apache Spark

- Apache Spark is a fast large-scale data processing engine and executes applications in Hadoop clusters 100 times faster in memory and 10 times faster on disk.

- Spark is built on data science and its concept makes data science effortless.

Excel

- Excel is a basic, popular and widely used analytical tool almost in all industries. Whether you are an expert in Sas, R or Tableau, you will still need to use Excel.

16. Define Statistics. [2 Marks]

Statistics is the study of the collection, analysis, interpretation, presentation, and organization of data.

17. Define statistical and non-statistical analysis. [2 Marks]

Statistical Analysis

- Analysis Based on statistical value.
- Providing complete insight for data.

Non- Statistical Analysis

- Based on generic information.
- Quantitative

18. List major categories of statistics. [2 Marks]

- There are two major categories of statistics: Descriptive analytics and inferential analytics.

19. Define Descriptive analytics and inferential analytics. [2 Marks]

- Descriptive analysis organizes the data and focuses on the main characteristics of the data.
- Inferential analytics uses the probability theory to arrive at a conclusion.

20. Write and explain steps for statistical analysis process. [5 Marks]



1. Defining the question

- The first step in any data analysis process is to define your objective. In data analytics jargon, this is sometimes called the 'problem statement'.

2. Collecting data.

Collect data from various sources from social media, machine, ecommerce website, excel files etc.

3. Cleaning data

- Removing major errors, duplicates
- Removing unwanted data points
- Filling in major gaps

4. Analysing data

- Different types of analysis (Descriptive, Diagnostic, Predictive, Prescriptive)

5. Sharing Result

- It involves interpreting the outcomes, and presenting them in a manner that's digestible for all types of audiences
- For this reason, data analysts commonly use reports, dashboards, and interactive visualizations to support their findings.